

Linux Filesystem Hierarchy

Directory-by-directory reference — what lives where, and why

A pure lookup reference for the standard Linux/RHEL top-level directory tree (FHS — Filesystem Hierarchy Standard). Every directory is documented with: what category of files it holds (config / binaries / logs / scripts / runtime / storage), the specific subpaths worth knowing, and whether it's commonly its own separate mount/partition. No labs, no tutorials — just fast cross-reference.

SCOPE: Based on the output of `ls /` on a RHEL system: `afs bin boot data dev etc home lib lib64 media misc mnt net opt proc root run sbin srv sys tmp usr var`. Every one of these is covered below, in the order they appear in that listing, plus a master lookup table up front for fast searching.

Master Quick-Lookup Table

"I'm looking for ___, where do I check?" — start here, then jump to the full entry for details.

I need...	Go to
System-wide config files (almost everything)	/etc/
A specific service's config (e.g. sshd, httpd)	/etc// e.g. /etc/ssh/sshd_config
Log files	/var/log/
A user's personal files & dotfiles	/home//
Root user's home directory	/root/
Everyday commands (ls, cp, bash)	/usr/bin/ (symlinked from /bin)
Admin-only commands (fdisk, mount, reboot)	/usr/sbin/ (symlinked from /sbin)
Shared libraries (.so files)	/usr/lib/, /usr/lib64/ (symlinked from /lib, /lib64)
Kernel, initramfs, bootloader files	/boot/
Device files (disks, terminals, null, etc.)	/dev/
Live kernel/process info (virtual, not on disk)	/proc/
Live kernel/hardware device tree (virtual)	/sys/
Runtime data since last boot (PID files, sockets)	/run/
Temporary files (cleared on reboot / periodically)	/tmp/
Manually mounted removable media (USB, CD)	/media/ or /mnt/
Third-party / manually installed software	/opt/
Data served to other systems (FTP, web, git)	/srv/
Variable/growing data (spool, cache, DBs, logs)	/var/
Package manager database & installed RPMs info	/var/lib/rpm/
Cron jobs (system-wide)	/etc/cron.d/, /etc/crontab, /var/spool/cron/
Systemd unit files (services)	/etc/systemd/system/, /usr/lib/systemd/system/
Environment variables set at login	/etc/profile, /etc/environment, ~/.bashrc
Network interface configs	/etc/NetworkManager/system-connections/
Persistent mount definitions	/etc/fstab
LVM metadata backups	/etc/lvm/backup/, /etc/lvm/archive/
SELinux config & policy	/etc/selinux/
firewalld rules	/etc/firewalld/
User account definitions	/etc/passwd, /etc/shadow, /etc/group

I need...	Go to
Sudo permissions	<code>/etc/sudoers, /etc/sudoers.d/</code>
Shell scripts you write for system tasks	<code>/usr/local/bin/</code> or <code>/opt/</code>
Autofs / automounter placeholders (usually empty)	<code>/afs/, /net/, /misc/</code>

1. Root and Core System Directories

/ — Root of the entire filesystem tree

The top of everything. Every other directory, disk, and mount point hangs off this single point. There is no "drive letter" concept in Linux — one unified tree.

Subpath	Holds
/ itself	Top-level directories only — rarely holds loose files directly

Always its own filesystem; typically a small LV (e.g. your rhel-root LV) mounted at boot before anything else.

/bin — Essential user command binaries

Core commands every user needs, even in single-user/rescue mode: ls, cp, mv, bash, cat, grep, etc.

Subpath	Holds
/bin/*	Executable binaries only — no configs, no scripts you'd edit

On RHEL 8/9/10 this is a SYMLINK to /usr/bin (the "usrmerge"). Not a separate mount.

NOTE: Because it's merged into /usr, you'll almost never interact with /bin directly — /usr/bin is the real location.

/boot — Kernel, initramfs, and bootloader files

Everything the system needs to boot BEFORE the root filesystem's LVM/normal storage is even active.

Subpath	Holds
/boot/vmlinuz-*	The compressed Linux kernel image, one per installed kernel version
/boot/initramfs-*.img	Initial RAM filesystem — drivers needed to mount root at boot
/boot/grub2/grub.cfg	GRUB2 bootloader configuration
/boot/loader/entries/	Boot Loader Spec (BLS) entries — one per kernel, used by grubby
/boot/efi/	EFI System Partition contents, on UEFI systems (often its own FAT32 mount)

ALWAYS its own separate partition (plain, non-LVM) — must be readable by the bootloader before LVM/kernel drivers load.

/data — Custom / user-created mount point (NOT part of standard FHS)

This is not a standard Linux directory — it exists on your system because you manually created it (mkdir /data) and mounted a partition or LV there for lab practice.

Subpath	Holds
/data/*	Whatever you've manually placed there — entirely user-defined

Whatever device you mounted with 'mount /dev/sdXN /data' or its /etc/fstab entry.

NOTE: Custom mount points like this are common in real deployments too (e.g. /data, /app, /backup) — they just aren't part of the FHS spec, so there's no fixed convention for what goes inside.

Every piece of hardware and many virtual devices are represented as special files here — this is how user space talks to disks, terminals, randomness sources, etc.

Subpath	Holds
/dev/sda, /dev/sdb...	Whole disk block devices
/dev/sda1, /dev/sda2...	Partition block devices
/dev/mapper/	Device-mapper nodes — LVM logical volumes, LUKS-encrypted volumes
/dev/null, /dev/zero	Special discard/source devices used constantly in shell scripting
/dev/tty*, /dev/pts/*	Terminal devices
/dev/random, /dev/urandom	Kernel entropy sources

A virtual filesystem (devtmpfs), populated dynamically by the kernel/udev at boot and on hotplug — not real disk storage.

2. /etc — Configuration Files (the most-referenced directory)

If you remember only one directory, make it this one. Virtually every system-wide config file for the OS and every installed service lives somewhere under /etc. It contains only text config files and small scripts — never binaries, never user data.

Subpath	Holds
/etc/fstab	Persistent mount definitions (filesystems + swap) read at boot
/etc/passwd, /etc/shadow, /etc/group	User accounts, password hashes, group memberships
/etc/sudoers, /etc/sudoers.d/	Sudo privilege rules
/etc/hosts, /etc/resolv.conf	Static hostname resolution, DNS resolver config
/etc/hostname	The system's hostname
/etc/NetworkManager/system-connections/	Saved network interface / connection profiles
/etc/ssh/sshd_config	SSH daemon configuration
/etc/systemd/system/	Custom/enabled systemd unit files (services, mounts, timers)
/etc/crontab, /etc/cron.d/, /etc/cron.daily/	Scheduled job definitions
/etc/selinux/config	SELinux mode (enforcing/permissive/disabled) and policy type
/etc/firewalld/	firewalld zones and rules
/etc/yum.repos.d/ or /etc/dnf/	Package repository definitions
/etc/lvm/lvm.conf, /etc/lvm/backup/, /etc/lvm/archive/	LVM tool config + automatic metadata backups
/etc/profile, /etc/bashrc, /etc/environment	Shell startup files and system-wide environment variables
/etc/skel/	Template files copied into new users' home directories on creation
/etc/sysctl.conf, /etc/sysctl.d/	Kernel runtime tuning parameters (e.g. vm.swappiness)

Usually part of the root (/) filesystem — not commonly its own separate mount/partition.

3. /home, /lib, /lib64

/home — Personal directories for every regular user

Each user gets a subdirectory here (e.g. /home/md) containing their personal files, downloads, and dotfile configs (~/.bashrc, ~/.ssh/, ~/.config/).

Subpath	Holds
/home//	That user's personal files
/home//.bashrc, .bash_profile	Per-user shell startup config
/home//.ssh/	Per-user SSH keys and known_hosts
/home//.config/	Per-user application configs (XDG standard)

VERY commonly its own separate mount/LV (as in your setup: rhel-home on /home) — keeps user data isolated from the OS.

/lib, /lib64 — Shared libraries needed by binaries in /bin and /sbin

The .so (shared object) files that executables link against at runtime — the C library, kernel modules, etc.

Subpath	Holds
/lib/modules//	Loadable kernel modules for each installed kernel
/lib64/*.so*	64-bit shared libraries

SYMLINK to /usr/lib and /usr/lib64 on RHEL 8/9/10 (usrmerge) — not a separate mount.

4. Removable Media & Automounter Directories

/media — Auto-mount point for removable media

Desktop environments and udisks automatically mount USB drives, CDs, and SD cards under here, typically namespaced by label or UUID.

Subpath	Holds
/media//	Auto-created directory per removable device, per user session

Not a persistent mount — created/removed dynamically as devices are plugged/unplugged.

/mnt — Conventional empty directory for TEMPORARY manual mounts

A traditional, purely conventional spot for sysadmins to manually mount something for a short-term task (rescue work, one-off inspection). Nothing auto-populates this.

Subpath	Holds
/mnt/*	Whatever you manually create/mount — empty by default

Empty by default; you create subdirectories and mount into them yourself as needed.

/afs, /net, /misc — Reserved mount points for the automounter (autofs)

Historically used by the autofs service to auto-mount AFS (Andrew File System) network shares and NFS exports on-demand when accessed. On most modern standalone systems these exist but stay completely empty because autofs / AFS isn't configured.

Subpath	Holds
/afs/*	Andrew File System mounts (if AFS client is configured — rare today)
/net/*	On-demand NFS auto-mounts (if autofs is configured)
/misc/*	Generic autofs mount point (if configured)

Not real storage until/unless autofs is configured and something is actively accessed under them.

NOTE: On your system these are almost certainly empty placeholders — safe to ignore unless you're specifically setting up autofs/NFS auto-mounting.

5. /opt, /proc, /root, /run

/opt — Optional / third-party software packages

Self-contained, manually-installed applications that don't follow the standard /usr layout — each vendor typically gets its own subdirectory.

Subpath	Holds
/opt///	A full self-contained install tree for one application

Part of / by default; sometimes given its own mount on servers hosting large third-party apps.

/proc — Virtual filesystem exposing live kernel & process info

Does NOT exist on disk at all — the kernel generates these files on-the-fly when you read them. This is how commands like ps, top, and free actually get their data.

Subpath	Holds
/proc//	Info about a specific running process (cmdline, status, open files)
/proc/cpuinfo, /proc/meminfo	Live hardware/memory info
/proc/mounts	Live, always-accurate list of currently mounted filesystems
/proc/swaps	Live list of active swap areas
/proc/sys/vm/swappiness	Live-readable/writable kernel tuning value

Virtual (procfs) — zero disk space used, resets completely on every boot.

/root — Home directory for the root user

Same purpose as a regular user's /home/, but for root — deliberately kept OUTSIDE /home so root can log in and have a working home dir even if /home fails to mount.

Subpath	Holds
/root/.bashrc, /root/.bash_profile	Root's shell startup config
/root/*	Root's personal files, scripts, downloaded install packages, etc.

Part of the root (/) filesystem by design — must be available even if /home doesn't mount.

/run — Runtime data since the last boot

Volatile storage for PID files, sockets, and state that's only valid for the current boot session. Replaced the older `/var/run` (which is now a symlink to `/run`).

Subpath	Holds
<code>/run/.pid</code>	Process ID files for running daemons
<code>/run/systemd/</code>	systemd's own runtime state
<code>/run/user//</code>	Per-user runtime files (session bus sockets, etc.)

tmpfs (RAM-backed) — wiped completely on every reboot, never touches disk.

6. /sbin, /srv, /sys, /tmp

/sbin — Essential system administration binaries

Like /bin, but for admin-only commands: fdisk, mkfs, reboot, iptables, parted, etc. — tools regular users typically don't run.

Subpath	Holds
/sbin/*	Admin executable binaries only

SYMLINK to /usr/sbin on RHEL 8/9/10 (usrmerge) — not a separate mount.

/srv — Data served to other systems

Meant for data your machine serves out via a protocol — FTP uploads, web content, git repos — kept separate from OS files.

Subpath	Holds
/srv/www/	Example: web server document root (convention, not enforced)
/srv/ftp/	Example: FTP server files

Rarely used by default on minimal installs; sometimes its own mount on dedicated file/web servers.

/sys — Virtual filesystem exposing the kernel's device/driver model

Like /proc, generated live by the kernel — this is the low-level view of every device, driver, and kernel subsystem, used heavily by udev and hardware-management tools.

Subpath	Holds
/sys/block/	One entry per block device known to the kernel
/sys/class/scsi_host/	SCSI host adapters — where you trigger disk rescans (echo > scan)
/sys/class/block/sda/device/rescan	Trigger a rescan of an existing disk's size (used earlier)
/sys/fs/	Filesystem-specific kernel tunables (cgroup, selinux, etc.)

Virtual (sysfs) — zero disk space, always reflects live kernel state.

/tmp — World-writable temporary storage

Any user or process can write here. Contents are expected to be short-lived — RHEL runs systemd-tmpfiles to periodically clean old files, and /tmp is often wiped or is tmpfs (RAM) entirely.

Subpath	Holds
/tmp/*	Short-lived temp files from any user/process — world-writable (sticky bit set)

Sometimes mounted as tmpfs (RAM-backed, cleared every reboot) — check with 'mount | grep /tmp'.

7. /usr and /var — the two biggest directories

/usr — Secondary hierarchy — almost the entire OS actually lives here

On modern RHEL (usrmerge), /usr is where essentially all installed software really resides. /bin, /sbin, /lib, /lib64 at the root are just symlinks pointing in here.

Subpath	Holds
/usr/bin/	Real location of nearly all user command binaries (/bin symlinks here)
/usr/sbin/	Real location of admin command binaries (/sbin symlinks here)
/usr/lib/, /usr/lib64/	Real location of shared libraries (/lib, /lib64 symlink here)
/usr/local/bin/, /usr/local/sbin/	Custom scripts/binaries YOU install, kept separate from packaged ones
/usr/share/	Architecture-independent shared data — man pages, docs, icons, default configs
/usr/share/doc/	Package documentation
/usr/lib/systemd/system/	Default systemd unit files shipped by installed packages
/usr/src/	Kernel source / headers, if installed

Part of / by default; on larger/older layouts sometimes its own dedicated mount.

/var — Variable data — anything that grows or changes at runtime

Everything here is expected to change size over time: logs grow, caches fill, spools queue up. This is why /var is frequently given its own mount/LV in production — so runaway log growth can't fill up and crash the root filesystem.

Subpath	Holds
/var/log/	System and service log files — messages, secure, audit/, httpd/, etc.
/var/lib/	Persistent state data for applications, e.g. /var/lib/rpm/ (package DB), /var/lib/mysql/
/var/spool/	Queued data awaiting processing — mail, print jobs, cron (/var/spool/cron/)
/var/cache/	Application cache data — safe to clear, regenerated as needed (e.g. dnf cache)
/var/tmp/	Temp files that should SURVIVE a reboot (unlike /tmp)
/var/www/	Common convention for web server document root on many distros
/var/run/	Symlink to /run (legacy path, kept for compatibility)
/var/crash/	Kernel crash dump storage (kdump)

Commonly its own separate mount/LV in production for exactly the reason above — isolates unpredictable growth from /.

8. One Structural Fact Worth Memorizing: usrmerge

On RHEL 8, 9, and 10 (and most modern distros), the historical split between `/bin` vs `/usr/bin` and `/lib` vs `/usr/lib` no longer exists as separate storage. It was merged years ago for simplicity. The symlinks below are why you'll sometimes see confusing outputs when checking file locations:

This path...	...is actually a symlink to
<code>/bin</code>	<code>/usr/bin</code>
<code>/sbin</code>	<code>/usr/sbin</code>
<code>/lib</code>	<code>/usr/lib</code>
<code>/lib64</code>	<code>/usr/lib64</code>

WHY THIS MATTERS: Practical effect: running `which ls` shows `/usr/bin/ls`, even though typing just `ls` works because `/bin` still resolves there via the symlink. You never need to "pick" the right one — they're the same file.

End of reference. Use the Master Quick-Lookup Table (page 3) as your fastest entry point next time you need a location.